

ROLLA, MO, USA

WMO: 724420

Lat: 38.132N

Lon: 91.765W

Elev: 1110

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 13997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.1	10.8	-3.4	4.8	7.6	1.1	6.1	13.1	26.0	42.5	24.0	40.1	8.2	310	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	94.7	75.2	91.4	75.0	88.9	74.4	78.5	88.9	77.3	87.6	75.9	85.6	8.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	138.7	85.1	74.1	132.7	83.3	72.9	127.2	81.7	42.8	88.9	41.4	87.4	40.1	85.8	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.8	18.0	-1.8	98.8	5.0	3.7	-5.4	101.5	-8.3	103.7	-11.1	105.7	-14.7	108.5
			-2.6	80.5	4.8	1.5	-6.0	81.5	-8.8	82.4	-11.5	83.2	-14.9	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.8	32.1	35.9	45.6	56.1	64.8	73.0	77.4	76.4	68.8	57.1	45.7	35.3
	DBStd	18.36	12.13	11.45	11.09	9.43	8.01	6.04	5.39	5.65	7.44	9.53	10.28	10.72
	HDD50	1952	566	413	217	45	4	0	0	0	40	198	469	
	HDD65	4680	1021	815	606	291	106	11	1	2	45	280	580	922
	CDD50	4069	11	19	80	229	462	690	850	820	566	260	70	12
	CDD65	1320	0	0	4	25	100	251	386	357	161	34	2	0
	CDH74	11796	0	2	31	205	740	2177	3814	3262	1326	227	12	0
	CDH80	4585	0	0	2	32	168	767	1697	1426	455	38	0	0
(14) Wind	WSAvg	9.0	10.3	10.3	10.6	10.7	8.7	7.8	6.9	7.0	7.3	8.8	9.9	10.0
(15) Precipitation	PrecAvg	37.7	1.7	1.6	3.4	3.7	4.6	4.1	3.5	3.5	3.3	3.2	3.2	2.4
	PrecMax	52.4	5.2	4.0	7.1	7.9	10.1	12.4	10.4	10.6	7.2	6.1	9.1	6.0
	PrecMin	28.5	0.2	0.3	0.7	1.1	1.7	0.1	0.9	0.7	0.5	0.2	0.7	0.4
	PrecStd	7.4	1.4	1.1	1.6	1.7	2.1	3.1	2.4	2.4	1.8	1.7	2.3	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	71.1	78.0	84.0	88.1	93.9	99.2	99.1	92.7	84.4	75.7	66.9
		MCWB	55.9	58.3	62.0	66.3	70.9	74.9	74.8	74.1	72.5	69.6	63.0	60.0
	2%	DB	61.0	64.3	72.9	79.6	84.1	90.0	94.7	95.1	89.0	80.0	70.0	61.3
		MCWB	53.2	55.0	59.5	63.4	71.0	75.6	75.7	75.3	72.8	67.0	58.3	55.3
	5%	DB	55.1	59.5	68.0	75.6	81.5	87.4	91.5	91.3	85.4	75.9	65.5	56.3
		MCWB	49.1	51.9	56.9	61.9	69.6	74.5	75.9	74.2	71.5	64.2	57.1	49.9
	10%	DB	49.9	54.2	63.4	72.0	78.6	84.5	89.0	88.0	81.7	72.0	61.6	51.5
		MCWB	43.8	47.4	54.4	60.2	67.6	73.1	75.4	73.6	69.5	62.8	54.0	45.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.8	61.1	64.7	68.9	75.2	79.1	80.3	80.0	76.5	72.5	65.1	61.8
		MCDB	65.1	68.2	74.6	78.9	83.6	88.4	90.2	90.9	87.7	81.3	72.0	66.6
	2%	WB	55.4	57.6	61.6	66.3	73.1	77.3	78.9	78.1	74.8	69.7	61.3	56.6
		MCDB	59.1	63.2	69.6	75.2	81.6	86.9	89.3	88.8	85.1	76.0	67.0	60.4
	5%	WB	49.8	52.9	59.0	64.2	71.0	75.6	77.6	76.5	73.2	67.2	58.7	51.5
		MCDB	54.9	57.9	66.5	72.7	79.0	84.4	88.2	86.8	82.4	72.8	64.3	55.1
	10%	WB	43.9	47.1	55.5	61.9	69.0	74.3	76.4	74.9	71.5	64.5	55.1	46.3
		MCDB	49.5	53.7	62.2	70.3	76.2	82.8	86.2	84.6	79.4	70.4	60.8	50.1
(35) Mean Daily Temperature Range	5% DB	MDBR	17.9	18.3	20.2	20.9	19.4	18.7	19.8	20.1	20.4	19.9	18.8	16.7
		MCDBR	24.3	26.1	25.0	24.4	21.5	20.8	22.6	24.0	22.5	22.1	22.9	22.9
		MCWBR	17.4	18.2	14.5	13.4	11.5	10.0	8.5	8.4	9.8	11.8	15.2	16.7
	5% WB	MCDBR	21.3	23.8	21.5	21.0	19.2	18.7	19.8	20.0	19.2	18.1	20.5	21.2
		MCWBR	17.4	18.6	14.1	13.2	11.3	10.1	8.9	8.8	9.2	11.2	15.5	18.0
(40) Clear-Sky Solar Irradiance	taub		0.297	0.309	0.335	0.381	0.417	0.424	0.449	0.448	0.385	0.342	0.307	0.294
	taud		2.488	2.449	2.430	2.306	2.261	2.272	2.218	2.227	2.378	2.490	2.536	2.520
	Ebn at Noon		282	293	294	285	275	272	264	261	273	277	277	276
	Edh at Noon		25	30	34	40	43	43	45	43	35	28	24	23
(44) All-Sky Solar Radiation	RadAvg		698	905	1214	1591	1791	2046	1994	1775	1528	1103	763	590
	RadStd		76	99	114	97	158	158	134	96	100	121	80	65

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (20 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137

Nomenclature: See separate page